

Forecast Agreement and Counterfactual Warrant in Adaptive Markets

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Abstract

Can a market become predictively stable without resolving how its rewards depend on contribution? This article argues that forecast agreement is an insufficient completion criterion for economic attribution. It separates definition, identification, learning, and decision, then uses equivalent complete-history laws to connect predictive merging with persistent posterior attribution uncertainty. An elementary two-state argument shows why positive prior odds continue to matter even when forecasts of the remaining future converge. A classical potential-outcomes example distinguishes this learning limit from the unidentified coupling underlying individual causal attribution. The methodological contribution is an evidence contract specifying the intervention, observation channels, model restrictions, adaptive sampling design, and decision stakes supporting a claim. It connects these obligations to debates about credible model worlds, explanatory validity, and organized criticism. Shadow histories and simulated intervention worlds remain scientifically useful when their model-to-world bridges are explicit. The policy implication is conditional: informative access can improve public decisions, but a comparison-budget limitation does not by itself establish a monetary externality or justify an antitrust intervention.

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JEL codes: B41, C18, D83, L41.

1. Introduction

A mature platform can predict its next transaction accurately while remaining unable to establish how much its observed advantage depends on productive contribution, inherited position, or a self-reinforcing allocation rule. This possibility matters because a market record performs several epistemic functions. Firms use it to forecast demand; investors use it to select projects; researchers use it to estimate mechanisms; public authorities use it to evaluate alternatives. Success at one function does not establish success at all the others. A record that supports an excellent forecast can be inadequate for a particular contribution claim.

This paper examines that distinction in markets that help determine the evidence subsequently available about themselves. Its central argument is that forecast agreement is an insufficient completion criterion for economic attribution. A completion criterion specifies when an inquiry has obtained the warrant appropriate to its question. When the question concerns an intervention or a contribution parameter, convergence of forecasts does not by itself discharge the obligations of definition, identification, and learning. The paper does not attribute this mistaken inference to a whole school of economics. It evaluates a tempting inference that can arise when predictive performance is used to validate an explanation or justify a market outcome.

Shadow Futures provides a precise setting for this argument (Erlic, 2026). Under its stated conditions, a finite comparison budget produces equivalent complete-history laws for different contribution parameters. Equivalence is weaker than equality: the laws may disagree on many event probabilities, yet share all probability-zero events. That overlap prevents a universally consistent estimator of a nonconstant parameter functional from separating the parameters through one growing history. At the same time, the classical predictive-merging theorem of

Blackwell and Dubins (1962) implies agreement about the conditional future. The same observation process can support predictive convergence while retaining attribution uncertainty. The contribution here is methodological rather than a new consistency theorem. It develops the consequences of this combination for the scientific status of counterfactual market claims, gives an explicit two-state posterior argument, and proposes a structured account of evidential warrant. The argument connects economic methodology on model worlds and explanatory validity with causal inference, Bayesian learning, and statistical decision theory. It also specifies when the epistemic case for preserving comparison becomes a policy argument and when it does not. No conclusion about the actual external cost of a named company follows from the illustrative models.

The discussion proceeds through four inferential obligations, an individual-attribution example, the adaptive-history result, and the posterior implications of equivalent laws. It then examines simulation, model-to-world inference, and the institutional organization of criticism. The resulting evidence contract makes an attribution claim inspectable without demanding that every useful decision wait for a unique causal explanation. Technical details appear in the appendix so that the methodological argument remains accessible independently of the formal probability notation.

2. Prediction, explanation, and the warrant appropriate to a question

Economic models often deliberately isolate mechanisms. Their empirical use therefore requires an account of how the features retained in a model support inference about a target economy. Sugden (2000) treats theoretical models as constructed worlds whose credibility matters for inference beyond those worlds. Mäki (2009) emphasizes the purposes and target systems involved in using models as surrogates. These approaches make the bridge to the target a

substantive obligation. A simulated alternative does not become evidence about an actual market merely because its internal dynamics are coherent.

Debates about false assumptions and explanation provide a related but distinct problem. Reiss (2012) formulates a tension between models' apparent explanatory achievements and their departures from reality. Hausman (2023) considers the structure and limits of economic reasoning with inexact claims. The present argument identifies a difficulty that survives even if the operative allocation model is correctly specified. A true model can generate an observation process that does not permit the desired attribution parameter to be recovered consistently. Model validity and the evidential sufficiency of a sampling design are separate questions.

Prediction is nevertheless a legitimate scientific and practical objective. Breiman (2001) distinguishes approaches organized around prediction from those organized around an assumed data-generating model. Mullainathan and Spiess (2017) discuss the different demands of prediction and parameter estimation in applied economics. Those distinctions should not be flattened into the proposition that predictive work lacks explanatory value. A successful prediction can test a model, guide a decision, and constrain a class of explanations. The narrower point is that the logical reach of that success depends on which explanations and parameters the forecast can distinguish.

Consider two economic descriptions that agree on near-certain future allocation to the incumbent but disagree about the responsiveness of allocation to verified input differences. A forecasting loss puts most weight on predicting the incumbent's next allocation. An attribution loss puts weight on estimating the input responsiveness. As reinforcement accumulates, the first loss can become easy to minimize while the second remains difficult. Calling the first achievement

evidential completion for the second silently substitutes a different target. This target substitution is the central methodological error examined here.

There is no requirement that a useful explanation recover every structural parameter. An explanation might establish a mechanism's presence, bound its effect, or identify a robust intervention despite uncertain parameter values. Its warrant should match that more limited claim. Conversely, a statement that market rewards measure individual contribution is stronger than a statement that reward forecasts are accurate. It needs a defined contribution functional and evidence that distinguishes the admissible ways of generating those rewards. The scientific issue is the alignment of the claim with the discriminating capacity of the experiment.

3. Four obligations in counterfactual attribution

The first obligation is definition. An attribution inquiry must specify an actor or population, an intervention, a horizon, and an outcome functional. Changing an input while preserving an allocation rule is different from changing access, retraining a model, or creating a new firm. Expected cumulative reward under an intervention is also different from the causal share of one realized reward. These quantities may be related, but they are not interchangeable. The intervention must make clear which resources, responses, and histories it is allowed to change. The second obligation is identification. Within an admissible model class, does the distribution of all the designated observations determine the target functional? If two models yield identical observation laws but different target values, additional observations from that unchanged law cannot point identify the target. Identification can improve through justified restrictions or through a different observation channel. Thus an identification statement is always relative to an information set and a class of models, even when the notation suppresses them.

The third obligation is learning under the sampling design. Identification from an entire probability distribution does not imply that one observed trajectory permits a consistent procedure at all parameter values. In an adaptive economy, observations are dependent and their informativeness changes with the history. Distinct complete-history laws can remain mutually absolutely continuous. They can then identify different distributions in a mathematical model while failing to generate the separating almost-sure events required by universal consistency. This is the specific role of the Shadow Futures impossibility theorem.

The fourth obligation is decision. A policymaker needs feasible actions, consequences, social weights, uncertainty treatment, and institutional authority. Statistical uncertainty can affect this decision without making action impossible. Manski (2013) develops decision analysis that retains uncertainty rather than requiring an unsupported point estimate. Equally, a well-estimated contribution parameter does not select a welfare objective or establish that an access intervention is beneficial. The inferential obligations therefore form a sequence of distinct questions, not a hierarchy in which complete attribution is a prerequisite for every decision.

This distinction also prevents an ambiguity in the term counterfactual. Potential outcomes organize questions about specified intervention regimes (Holland, 1986; Hernán and Robins, 2020). Structural models supply another explicit way to describe interventions and their invariances (Pearl, 2009). Interventionist explanation concerns the appropriate stability of relations under changes (Woodward, 2004). One can accept these practical inferential distinctions without settling every philosophical question about the ontology of unobserved potential outcomes. What matters here is whether the chosen representation supplies a coherent intervention and an evidential route to the stated target.

4. Individual attribution and the missing coupling

A simple example isolates an identification problem that must be kept separate from the adaptive-history learning result. Let treatment A be randomized independently of a binary variable U , with U equally likely to be zero or one. Each unit reveals only its realized outcome $Y=Y(A)$. Suppose both treatment values have positive probability. Two possible couplings of the potential outcomes are:

$$\text{Model I: } Y(0) = U, Y(1) = U. \quad (1)$$

$$\text{Model II: } Y(0) = U, Y(1) = 1 - U. \quad (2)$$

Proposition 1. Models I and II have the same joint distribution of the observed pair (A, Y) , and the same outcome marginal under either intervention setting A . They have different distributions of the individual effect $Y(1) - Y(0)$.

For either treatment value, the outcome is Bernoulli with probability one-half in both models. Randomization therefore gives identical observed laws and identifies an average treatment effect of zero. In Model I every individual effect is zero; in Model II it is $+1$ or -1 with equal probability. The intervention marginals do not determine how the two unrealized outcomes are coupled within units. This is a classical causal-inference distinction, presented to discipline the attribution target rather than to claim a new identification result.

An economic inquiry into the average effect of an input can consequently succeed while an inquiry assigning individual credit remains unresolved. The distinction is relevant to statements that one project's realized reward represents its causal contribution. Even a randomized allocation of opportunities need not identify every individual counterfactual assignment. Additional structural assumptions might do so, but their role must be explicit. Examples include restrictions on outcome monotonicity or a credible repeated-unit intervention with stable outcomes and no carryover.

This example does not establish that causal effects are generally unknowable. The average effect is identified here, and the randomized design can estimate it consistently using independent units. It shows instead why the phrase contribution uncertainty needs a named estimand. A theorem about a distribution of individual effects cannot automatically be invoked against a population average; a theorem about recovery of an input-loading parameter cannot automatically be invoked against an observable forecast. Careful target definition avoids moving an impossibility result to a problem it does not address.

5. Endogenous evidence and equivalent histories

In a fixed-recipient allocation model, let $p(i,t)$ denote the conditional probability that recipient i receives allocation t given the raw past. The comparison budget sums probability outside the most likely recipient. It is an object of the specified experiment: it concerns the conditional opportunities produced by its allocation rule, not a physical stock of knowledge that can be added across arbitrary markets without further assumptions.

$$\varepsilon_t = 1 - \max_i p(i, t), \quad B(T) = \sum_{t < T} \varepsilon_t. \quad (3)$$

Under bounded profiles in a conditional-logit allocation model, the conditional covariance of the selected input profile is the Fisher information of the next allocation. In the scalar case, V5 bounds that variance by a constant times probability outside the leader. A large transaction count can therefore coexist with a small accumulation of input-discriminating information. The bound is directional when the parameter is multidimensional: variation can be present in one direction and absent in another. Neither a scalar comparison measure nor a scalar transaction count substitutes for the complete information matrix.

The complete-history impossibility result needs more than a small observed budget at a finite horizon. V5 assumes the canonical recipient-history filtration, common predictable profiles and

state rules, a fixed initial history, positive finite-history kernels, and a Hellinger-distance bound tied to comparison. Almost-sure finiteness of the budget then yields equivalence of the fixed-parameter complete-history laws. Its non-estimation conclusion concerns procedures based on that experiment, with pointwise consistency required at every parameter. It is not a numerical lower bound on every estimator's accuracy in every realized sample.

The distinction between equality and equivalence is crucial. Equality creates the identification failure illustrated by Proposition 1. Equivalence permits distinct probabilities and hence finite-sample evidence in favor of one parameter, but prevents the infinite record from almost surely separating the parameter laws. One can learn something without learning enough for universal consistency. Describing equivalence as complete ignorance would discard the ordinary likelihood evidence that remains available.

Related work already establishes incomplete learning when actions determine the data collected. Brezzi and Lai (2000), for example, analyze incomplete learning from endogenous data in dynamic allocation. The general theme is therefore not new. The comparison-budget program's candidate contribution is a tractable object and sufficient conditions in a specified reinforced experiment. This paper's methodological contribution concerns how that result should alter the standards for attributing rewards or extrapolating from forecast performance, rather than relabeling all endogenous-learning failures as shadow futures.

The observation boundary must be defended empirically. A recommender may observe known exposure decisions alongside uncertain click responses, purchases, complaints, and outside experiments. These are different factors in a joint likelihood. Response data can supply information even when recipient allocation becomes concentrated. Likewise, independent market replications with a shared structural parameter can create a separating design. A claim about an

allocation-only history must not suppress informative channels that the actual analyst or institution can access.

6. Forecast merging with persistent attribution uncertainty

Let P_0 and P_1 be distinct but equivalent laws of a complete history, and let the increasing information sets eventually generate the full history information. Consider a Bayesian inquiry whose two possible parameter states induce these laws and whose prior gives both states positive probability. The following elementary consequence makes the distinction between forecasting and attribution particularly transparent.

Proposition 2. Under either fixed history law, the two-state posterior converges almost surely to a value strictly between zero and one. Its limiting odds equal the prior odds multiplied by the finite, positive complete-history likelihood ratio. Meanwhile, the conditional forecasts of the entire remaining history under P_0 and P_1 merge in total variation, almost surely under either law. The posterior statement follows from the likelihood-ratio martingale; the forecast statement is an application of Blackwell and Dubins (1962). The appendix gives the posterior derivation. Neither component is offered as a new Bayesian theorem. Together they show that an investigator can face asymptotically indistinguishable future forecasts while retaining a substantive posterior dependence on the prior attribution odds. Different investigators observing the same complete history retain different limiting state odds if they began with different positive odds.

$$\frac{\pi_\infty}{1 - \pi_\infty} = \frac{w_1}{w_0} Z_\infty, \quad 0 < Z_\infty < \infty. \quad (4)$$

Strictly interior posteriors need not be close to one-half. A likelihood ratio can strongly favor one state, and a decision can be effectively settled despite the lack of complete parameter recovery.

Equivalence supplies no uniform quantitative lower bound over all equivalent pairs. The residual uncertainty may be economically negligible or consequential depending on the loss function and

prior. This is why a mathematical impossibility result must not be converted directly into a monetary externality.

There is, nevertheless, a positive minimum complete-history Bayes error for any fixed equivalent two-state pair with positive prior weights. Under zero-one state-classification loss, the overlap integral is strictly positive. Acquiring the entire realized history cannot reduce this particular error to zero. This supplies a well-defined residual evidential limitation rather than the imprecise statement that contribution is impossible to know. It also distinguishes the claim from a mere problem of insufficient computing capacity: exact likelihood calculation cannot create separating observations.

A second boundary concerns mixtures. Two equivalent prior-predictive distributions can produce merging forecasts even if the fixed-state components are mutually singular and the state is learnable. Predictive merging by itself therefore does not diagnose a comparison-budget obstruction. The inference in Proposition 2 requires equivalence of the particular fixed-parameter history laws. Bayesian consistency also has separate complications in larger parameter spaces, including those studied by Diaconis and Freedman (1986). They should not be treated as the same mechanism as the elementary fixed-pair overlap analyzed here.

The methodological implication is conditional and precise. When equivalent fixed-state laws are established for the relevant observation design, forecast agreement cannot certify that the attribution inquiry has eliminated prior-sensitive state uncertainty. In other designs, forecast agreement may accompany successful parameter learning. The appropriate evidential standard must therefore inspect the experiment and the target, rather than infer an attribution achievement from convergence alone.

7. Shadow histories, intervention worlds, and simulation

A shadow history in the foundational paper is a possible finite-horizon allocation path generated under the same structural parameter, initial state, and predictable rules. It describes an unrealized branch of that experiment. An intervention world changes specified features of the experiment. A proposed counterfactual company additionally requires a feasible production technology, resource envelope, access arrangement, and service identity. These three objects should be distinguished before asking what evidence establishes them.

Enumerating model-generated histories answers questions about that model. It can expose path dependence, calculate a likelihood, compare designs, or show that an estimator fails under known truth. It cannot by itself establish which parameter generated a historical economy.

Generating one million paths from an assumed loading adds computational precision to a conditional implication; it does not create one million independent observations about the actual loading. The relevant empirical warrant comes from validating the generator and testing alternatives with observations not determined by the same untested assumptions.

Credibility therefore has at least two components. A model world must possess a coherent economic interpretation, and its discriminating implications must connect to observable or experimentally accessible features of the target. A plausible story about a suppressed competitor addresses the first component only partially. A resource-constrained prototype, benchmarked task, and specified access change can turn that story into an investigable hypothesis. Failure of the prototype might refute an optimistic claim; success can support a bounded capability statement while leaving demand, scale, and equilibrium responses uncertain.

Transport is another independent bridge. Cartwright and Hardie (2012) emphasize why evidence that a policy worked in one setting does not settle whether it will work elsewhere. In the present

setting, an independent procurement trial can identify performance under the trial's task distribution while failing to identify long-run market entry under a platform's state dynamics. The direction of that gap should be stated rather than hidden. A narrowly successful experiment can still be valuable if the actual public decision concerns the narrow task it measures. One practical consequence is to separate computation used for demonstration from computation used for empirical inference. Synthetic experiments should declare their generating parameters, seeds, horizons, observation channels, and known truths. Empirical counterfactual exercises should declare which restrictions come from external evidence and which are maintained assumptions. A reproducible calculation makes the conditional implication inspectable. It strengthens, but does not replace, the model-to-world bridge.

8. An evidence contract and organized criticism

An evidence contract is a proposed public specification of what an attribution claim means and what supports it. It comprises the target and intervention; the observation channels and their selection rules; the admissible model class; the identification and learning conditions; and the decision for which residual uncertainty matters. Calling it a contract emphasizes inspectability and the possibility of challenge, rather than a guarantee that every target can be recovered. It can accompany an article, policy evaluation, or procurement protocol.

For an adaptive experiment, the contract should identify the raw information actually recorded. A state summary is sufficient only if its sufficiency is established for the chosen target. Recording a provider's cumulative count may omit order-dependent responses or changing alternatives. The contract must also state which functions are held invariant across parameter comparisons. Common predictable rules mean the same realized history maps to the same profiles and states;

they do not mean that different histories generate identical realized inputs. This distinction is essential when assessing a history-law equivalence claim.

For causal use, it should state the additional invariances connecting an observational parameter to the intervention. Changing access may alter investment, adaptation, demand, and the meaning of a service. Holding these responses fixed can define an informative controlled effect, but that effect is different from the full equilibrium consequence of the access policy. A controlled estimate should not be advertised as a comprehensive welfare effect. The broader effect requires its own model, uncertainty set, or empirical design.

The contract should then distinguish mathematical implications, statistical estimates, simulated results, and normative commitments. A Lean proof verifies a formal proposition with its encoded assumptions. It does not establish that a market satisfies those assumptions or choose a social loss function. Conventional mathematical proof remains the scientific standard unless a venue requires formal verification. Reproducible computation demonstrates what an implemented procedure does under declared conditions; it does not transform those conditions into measured facts.

Organized criticism matters because no observation boundary or background assumption is automatically neutral. Longino (1990) connects scientific objectivity with social practices of criticism, while Douglas (2009) analyzes the legitimate role of values under uncertainty. These perspectives support making alternative models and decision consequences open to scrutiny. They do not imply that adding firms mechanically produces scientific objectivity. A second provider that shares the same hidden ranking process may add little independent evidence; a single institution with protected, independent tests can add more.

The proposed standard is therefore procedural as well as analytical. Qualified critics need enough information to challenge the model, reconstruct the relevant experiment, and identify alternative observation designs. Disclosure should remain proportionate to privacy, security, intellectual-property, and implementation constraints. Independence is a property of the evidence-producing procedure, not a virtue automatically conferred by organizational multiplicity. A credible evidential institution protects meaningful opportunities to contest a claim while accounting for the cost of doing so.

8.1. A worked procurement evidence contract

Consider a hypothetical public organization choosing an AI-assisted document-review service for a registered class of tasks next quarter. Both an existing system and a qualified alternative satisfy the same minimum security, latency, compute, and human-oversight requirements. The bounded performance claim is: replacing the existing system with the alternative lowers expected review-adjusted loss for this task population under those requirements. It concerns a defined service decision. No trial has been conducted here and no supplier performance value is asserted.

The contract first defines the contrast. Each task has a potential loss under either system, including preregistered weights for error, review time, latency, and operating expense. The target is the population mean loss under the alternative minus that under the existing system. Tasks are sampled from the registered population and assigned with known positive probabilities to one system. User or workflow clusters are the assignment units where carryover or shared context would otherwise contaminate a task-level contrast. The record includes eligibility, task covariates, assignment probabilities, system versions, resource limits, outputs, failures, blinded scoring, review time, and costs.

Identification requires the assigned service to match its specified intervention, valid sampling or explicit reweighting to the task population, and a defensible interference restriction. Random assignment identifies the two mean losses under these conditions; it need not identify the joint distribution of individual task effects. Dependence within assignment clusters and multiple performance endpoints must enter uncertainty calibration. The routing probabilities are known features of this experiment. Their allocation score supplies no information about an unknown platform loading; the performance information comes from recorded responses.

The procurement decision uses the loss contrast and full transition cost. Adopt the alternative only when a registered uncertainty bound implies that aggregate expected loss savings over the intended task volume exceed migration, training, verification, and other additional costs, while mandatory reliability constraints remain satisfied. Otherwise retain the existing system or fund a separately justified follow-up. Applying the trial to next quarter requires stable versions, resource conditions, scoring, and a justified relationship between trial and deployment task distributions. A change in these conditions changes the target or adds a transport assumption.

Now compare the claim that the incumbent's observed market revenue measures its causal contribution. This claim concerns a different outcome, population, mechanism, and intervention. A contribution inquiry might replace the incumbent with a qualified alternative under a bounded resource envelope and measure the resulting welfare difference. It must specify how prices, entry, investment, user demand, and retained allocation state respond. A procurement trial holding these objects fixed does not measure that equilibrium contrast. Nor does its randomized task routing identify the input-loading parameter of an observed platform's reinforcement kernel. Estimating that loading requires the relevant chronological platform records and a justified mechanism class.

The evidence contract therefore records a change of question when a bounded trial is invoked to explain market rewards. Strong task performance can warrant adoption while leaving reward attribution unresolved; poor performance can reject this procurement alternative without proving that every potential entrant lacks contribution. Critics can challenge sampling, versions, scoring weights, or transport directly for the first claim. They must challenge the counterfactual replacement and market-response assumptions for the second. Table 1 makes those different evidential obligations visible without inventing empirical results.

Table 1. Two claims and their evidence contracts

Obligation	Bounded service-performance claim	Reward-as-contribution claim
Target	Mean review-adjusted task loss difference	Declared welfare difference under feasible replacement
Observation	Registered randomized task and response records	Chronological market records and relevant external channels
Intervention	System replacement at fixed service requirements	Replacement with stated market and state responses
Learning	Valid task-population uncertainty calibration	Identification and learning for the declared mechanism
Decision	Procurement savings versus full transition cost	Separate policy objective, feasible instrument and authority

Hypothetical design; the trial alone neither identifies platform reinforcement nor attributes observed revenue.

9. Decisions under unresolved attribution

Residual attribution uncertainty can have different policy consequences. It may be harmless if every admissible model recommends the same action. It may justify a limited experiment if the expected improvement in a consequential decision exceeds the experiment's cost. It may support a robust remedy when a common feasible rule performs acceptably across models. Or it may

counsel against intervention when productive losses are large and evidential gains small. A failure of point recovery selects none of these outcomes by itself.

Statistical decision theory makes this distinction explicit. Blackwell (1953) compares experiments through their usefulness across decision problems. The relevant gain is a reduction in expected loss for a specified decision, not simply an increase in a scalar data quantity. A comparison-budget bound can help diagnose a limitation of an allocation experiment, but the budget has no intrinsic exchange rate into money. Different decision stakes, priors, nuisance mechanisms, and response channels can attach different values to the same budget.

An epistemic competition argument consequently needs three additional links. A feasible access arrangement must supply genuinely discriminating evidence. That evidence must improve a decision with a declared social objective. Its benefits must exceed productive, administrative, privacy, and other costs. An externality claim adds a fourth link: some of the net effect must fall outside the operator's private incentive. Only after these links are established does corrective taxation become an instrument question rather than a metaphor about paying for unrealized possibilities.

This approach can evaluate a bounded access intervention without treating concentration as its economic primitive. It still requires a boundary describing the service, affected decisions, qualifying alternatives, resource constraints, and plausible substitutions. Existing legal requirements about market definition or dominance remain constraints on actual antitrust enforcement. Epistemic analysis can inform the evidence and the design of other lawful instruments; it cannot waive statutory predicates by changing the vocabulary of the economic argument.

For AI systems, a defensible starting point is a specified deployment decision rather than a general verdict on hyperscaling. Independent trials may reveal whether a feasible alternative meets a public task's performance, reliability, and cost requirements. The results can affect procurement, evaluation access, or investment decisions. They do not establish the optimal tax burden of every provider. Scaling benefits can be real at the same time as an access architecture creates a modifiable evidential cost; both enter the comparison.

10. Objections and limits

One objection is that real economies offer more information than allocation histories. This is often correct and is a reason to specify the complete observation experiment. The argument survives only where equivalent laws remain after the relevant channels are included, or where finite-horizon discrimination is inadequate for the decision at issue. Additional response data, independent replications, or interventions can remove the premise of the impossibility result. An epistemic diagnosis should actively seek those channels rather than protect a restrictive model from refutation.

A second objection is that a universally inconsistent estimation problem can still permit a good estimate at the actual parameter. That is also correct. The theorem rules out one procedure working consistently at every parameter in a nonconstant target; it does not classify every realized estimate as worthless. Posterior evidence, confidence regions, and decision losses remain meaningful within their stated guarantees. A claim of universal recovery is stronger than a claim of local usefulness, and the manuscript preserves that difference.

A third objection is that all science uses assumptions, so an evidence contract adds nothing. Its value is not the discovery that assumptions exist. It places the intervention, observation boundary, identification restrictions, adaptive sampling conditions, and decision stakes together,

making visible which bridge each assumption supplies. In particular, it prevents model fit from silently doing the work of intervention invariance and prevents forecast merging from silently doing the work of attribution recovery. These are separable mistakes with different repairs.

A fourth objection concerns the politics of evidence production. Public evaluation can be captured, expensive, or used to favor weak alternatives. This objection weakens any automatic preference for intervention. It strengthens the case for bounded eligibility rules, auditable protocols, independent analysis, and stopping criteria. An epistemic argument is professionally useful when it can recommend against a trial whose expected value is negative, not only when it supports a favored competitor.

Finally, the argument is not a general impossibility of counterfactual science. It identifies specific ways in which a realized adaptive process can fail to answer an attribution question. Other designs may answer it, and some decisions do not require its answer. The constructive response is to locate the missing discriminating experiment and assess whether it is feasible and worth producing. This retains the scientific role of counterfactual reasoning while declining to treat imagined alternatives as established economic losses.

11. Conclusion

Forecast agreement and attribution warrant are distinct achievements. Under equivalent fixed-parameter history laws, investigators can converge on predictions of the remaining future while retaining positive complete-history classification error and prior-sensitive state odds. This classical probabilistic combination gives a concrete methodological limit on using predictive convergence as the completion criterion for a contribution inquiry.

Economic institutions affect the observation designs through which such inquiries proceed. The proposed evidence contract makes those designs, their intervention bridges, and their decision

uses explicit. It supports stronger counterfactual science by clarifying what would count as discriminating evidence. Its policy implication is a conditional program of informative access and evaluation: create or preserve a feasible experiment when its decision value justifies its full cost, and report the remaining uncertainty when it does not.

Appendix A. Posterior overlap and the scope of the propositions

A.1. Let F denote the full history information and let F_t increase to F . Suppose P_0 and P_1 are equivalent probability laws on F . The Radon–Nikodym derivative $Z_\infty = dP_1/dP_0$ exists, is finite and strictly positive P_0 -almost surely, and has P_0 -expectation one. The finite-history derivative is $Z_t = E_0[Z_\infty | F_t]$. Martingale convergence for conditional expectations gives $Z_t \rightarrow Z_\infty$ almost surely and in L^1 under P_0 . Equivalence transfers the almost-sure conclusion to P_1 .

A.2. With positive prior weights w_0 and w_1 summing to one, Bayes' formula gives $\pi_t = w_1 Z_t / (w_0 + w_1 Z_t)$. Continuity and positivity yield $\pi_\infty = w_1 Z_\infty / (w_0 + w_1 Z_\infty)$, which lies strictly between zero and one under either law. Two investigators with different positive prior odds see posterior odds whose ratio remains the ratio of their prior odds at every horizon and in the limit. This result is for a common, correctly specified two-state experiment; it is not a statement about arbitrary misspecified Bayesian procedures.

A.3. Put $M = P_0 + P_1$ and let f_0 and f_1 be the respective densities with respect to M . The optimal zero-one classification risk after observing the full history is the integral of $\min(w_0 f_0, w_1 f_1)$ against M . Equivalence makes both densities positive M -almost everywhere, so the integrand is positive almost everywhere and the integral is strictly positive. Finite-history Bayes risks are no smaller because their decision rules use a smaller information set. Across varying equivalent pairs this positive number can approach zero, so the proposition supplies no universal numerical floor.

A.4. Blackwell–Dubins merging applies because the two complete-history laws are mutually absolutely continuous. On the canonical countable-alphabet history space, their conditional distributions of the entire remaining history merge in total variation almost surely under either law. Applying the result to each fixed pair does not assert convergence uniform over an uncountable parameter class. It also does not equate the fixed-state laws with prior-predictive mixtures. Those distinctions are necessary for the methodological inference in Section 6.

A.5. In Proposition 1, observing intervention marginals gives only the two marginal Bernoulli distributions of $Y(0)$ and $Y(1)$. Independence of treatment and U is enough to identify their means when both treatments occur. It does not identify the cross-world coupling. The two couplings differ despite identical observed and marginal-intervention laws. In the adaptive-history result, by contrast, the laws are generally distinct but equivalent. The former is an identification obstruction; the latter is a learning obstruction under a specified design. They are presented together to separate, not merge, their inferential roles.

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Declarations

This article contains theoretical and methodological analysis and no human-subject or proprietary empirical data. The illustrative potential-outcomes construction and posterior arguments are fully specified in the text. Substantive AI assistance through OpenAI Codex was used to develop and revise prose, mathematical arguments, reference checks, and document production. The author is responsible for the manuscript.